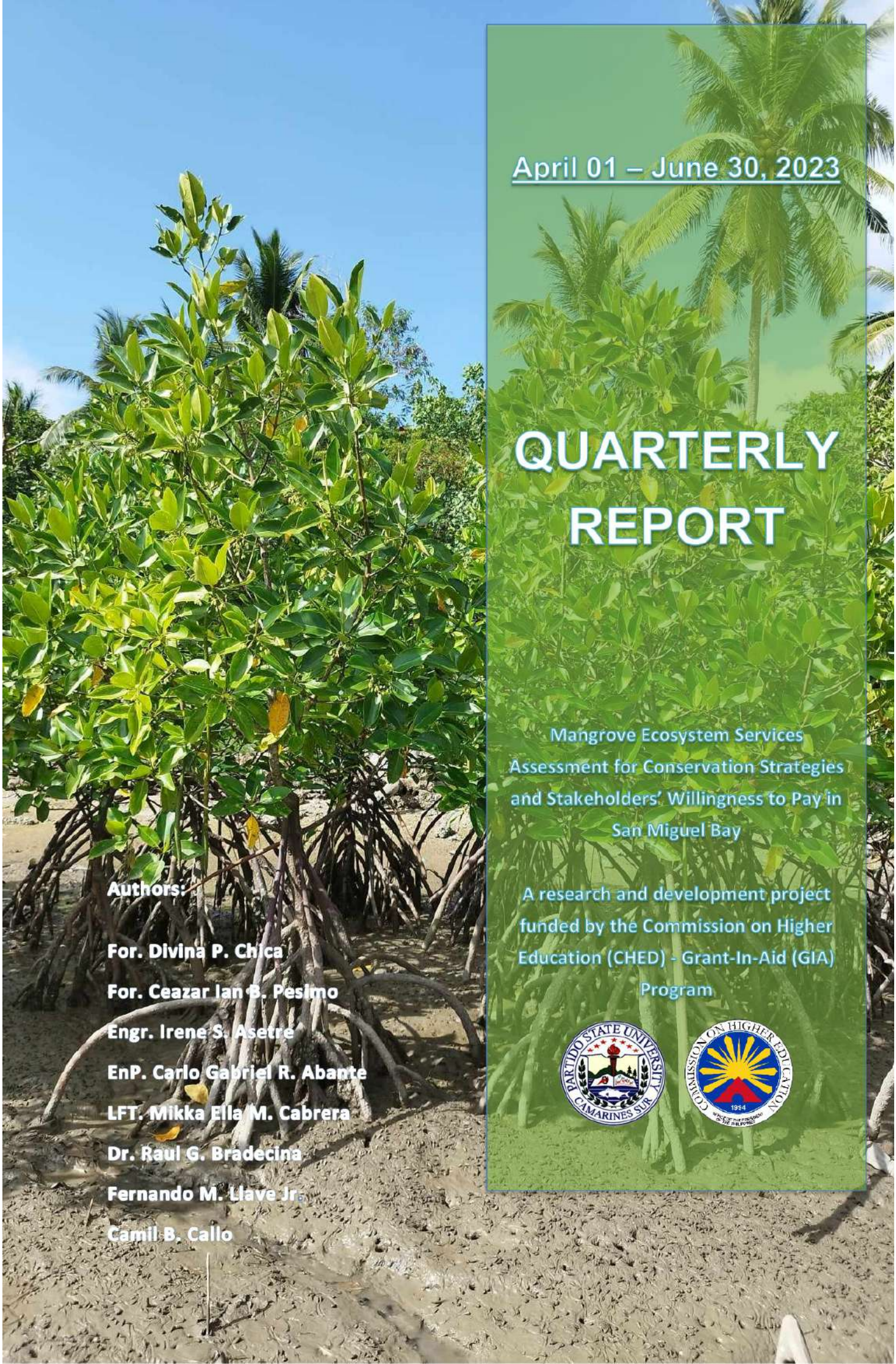


April 01 – June 30, 2023

QUARTERLY REPORT

Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay

A research and development project funded by the Commission on Higher Education (CHED) - Grant-In-Aid (GIA) Program

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About the Project

Project Title:

Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay

Component Studies

1. Community Structure and biodiversity Assessment of Mangrove Ecosystem in San Miguel Bay
2. Inventory of Ecosystem Services of Mangrove Forest and Narrative of the Benefits to Coastal Communities of San Miguel Bay
3. Assessment of Fisheries Production of Mangrove Ecosystem in San Miguel Bay
4. Contingent Valuation Study on Mangrove Forest Ecosystem Conservation and the fisher folks and coastal communities in San Miguel Bay
5. Socio-psychological Factors affecting Utilization and Willingness to Participate in the Management and Conservation of Mangrove Ecosystem of San Miguel Bay

Project Members

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Project Activities

This quarterly report provides a comprehensive overview of the research activities conducted by the research team during the period from April to June 2023. It aims to summarize the key findings, progress, and outcomes achieved throughout the specified timeframe. This report facilitates effective communication, evaluation, and strategic planning for future endeavors.

In this case, the project activities for the project as scheduled are the following:

- Implementation Progress
- Administrative Support Activities
    - Preparation of materials and equipment for fieldwork activities
    - Status of Memorandum of Agreements (MOA) with seven local government units (LGUs) surrounding San Miguel Bay (SMB); and securing Municipal Clearance
    - Securing of Prior Informed Clearance and Gratuitous Permit for Cabusao Wetland Critical Habitat and Mercedes Critical Habitat
    - Preparation of Quarterly Report (Q2)
- Research Undertakings
    - Questionnaires
    - Ocular Visit/Site Validation
    - Enumerator's Orientation and Training
    - Mangrove Forest Inventory
    - Fisheries Assessment

Implementation Progress

The implementation progress of the research project has been steadily advancing during the reporting period.

Administrative Support Activities

The administrative support activities in this project, for the period of April to June 2023, includes the following: (1) preparation of materials and equipments for fieldwork, (2) updating the status of Memorandum of Agreements (MOA) with seven localities surrounding SMB and securing of Municipal Clearances, Prior Informed Clearances and Gratuitous Permit for Cabusao Wetland Critical Habitat and Mercedes Critical Habitat; and (3) preparation of Quarterly Report (Q2).

During the 2nd Quarter of implementation of this project, the project staff and project support staff worked together to prepare the materials and equipment for fieldwork activities. This includes the preparation of survey questionnaires for enumerator's orientation and training, maps, mangrove tags and other field equipment for mangrove forest inventory and fisheries assessment.

In terms of necessary documents, project support staff facilitated the communication with seven LGUs regarding the status of MOAs, municipal clearances and permits such as prior informed clearance and gratuitous permit.

By the end of the 2nd quarter, reports have been prepared summarizing the overall conducted activities.

Research Undertakings

In every research task, achieving efficiency relies on proper preparation and the adoption of a methodical approach. Starting April 1, 2023 until June 30, 2023, the following are the tasks which are accomplished by the project team, relative to the approved timeline of the project i.e., survey questionnaires; ocular visit/site validation; and mangrove forest inventory and fisheries assessment. The following sections will be discussing the details of the mentioned activities, including photo documentation and research outputs.

Memorandum of Agreement (MOA), Clearances, and Permits

Table 1. shows the status of MOA, clearances and permits. Three (3) out of seven (7) LGUs have issued an SB resolution for MOA. In terms of municipal clearance, three (3) out of seven (7) LGUs issued a clearance (**Appendix**) while two (2) requested prior informed clearance for critical habitats were secured (**Appendix**).

Table 1. Status of MOA, Clearances, and Permits

LGU	MOA	Clearance	Permit	Remarks
Siruma	*For SB resolution*	*On process*	N/A	
Tinambac	*For SB resolution*	*On process*	N/A	
Calabanga	*For SB resolution*	✓	N/A	
Sipocot	*Issued SB resolution*	*On process*	N/A	• SB Resolution have been issued and ready for MOA signing • The LGU will only release a clearance once the MOA is signed.
Cabusao	*For SB resolution*	✓	Prior Informed Clearance: ✓ Gratuitous Permit: ✓	
Basud	*Issued SB resolution*	*On process*	N/A	• SB Resolution have been issued and ready for MOA signing

Mercedes	*Issued SB resolution*	✓	Prior Informed Clearance: ✓	• SB Resolution have been issued and ready for MOA signing • Submitted an application for Gratuitous Permit for Mercedes Critical Habitat to DENR R5

Questionnaires

The survey questionnaires for Components 2, 3, 4 and 5 were improved and finished. Project team has prepared the necessary documents to be submitted to the University Ethics Committee for review to secure the ethics clearance.

During the orientation and training of enumerators, these questionnaires were used and validated on the field.

Ocular Visit/Site Validation

On May 9 - 17 2023, the Project Team conducted an ocular visit and site validation in seven (7) municipalities along San Miguel Bay. This aims to gather preliminary data on location of mangrove areas as preparation for the incoming mangrove and fisheries assessment. Moreover, geotagging and photodocumentation was done in the field.

Photo Documentation





Figure 1. Project Team from Partido State University - Tinambac Campus, coordinating with the local government unit personnel, during the validation of local map for study sites and ocular visit in mangrove areas.

Enumerator's Training and Orientation

On May 9 - 17 2023, after the Project Team conducted an ocular visit and site validation in mangrove areas, training and orientation for enumerators took place. The enumerators were oriented on survey questionnaires, household surveys and their monthly targets. Additionally, pilot testing was undertaken on field.

Photo Documentation



Figure 2. Enumerator orientation and pilot testing of questionnaires in seven (7) LGUs

Mangrove Forest Inventory

The Mangrove Forest Inventory conducted during the Second Quarter of Year 2023. Below is the summarized table of mangrove assessment in seven LGUs.

Table 2 shows the summarized mangrove assessment in seven (7) LGUs. The assessment was conducted from the month of May to June 2023 in nine (9) Stations covering eight (8) barangays. Ten (10) transects with fifteen (15) plots were established and assessed.

Table 2. Second Quarter Mangrove Assessment

Local Government Unit	**Date**	**Barangay**	**Station**	**Transects/Plots**
Siruma	May 25, 2023	Poblacion	S90	T1 - P1,P2
Tinambac	May 22, 2023	Old Caaluan	S53	T1 - P1,P2
Calabanga	May 29, 2023	San Lucas	S38	T1 - P1,P2,P3
Cabusao	May 24, 2023 June 23, 2023	Barceloneta	S24 S25	T1 - P1 T1 - P1
Sipocot	May 23, 2023 June 23, 2023 June 22, 2023	Mangga Cotmo	S23 S20	T1 - P1 T1 - P2,P3 T2 - P1
Basud	May 31, 2023	Mangcamagong	S5	T1 - P1
Mercedes	May 25, 2023	Matoogtoog	S8	T1 - P1

Table 3 shows the initial data on mangrove forest inventory conducted in seven (7) LGUs. There are thirteen (13) mangrove species identified, namely, *Avicennia rumphiana*, *Avicennia marina*, *Avicennia officinalis*, *Bruguiera cylindrica*, *Bruguiera parviflora*, *Ceriops decandra*, *Excoecaria agallocha*, *Rhizophora mucronata*, *Rhizophora apiculata*, *Rhizophora stylosa*, *Sonneratia alba*, *Sonneratia ovata*, and *Xylocarpus granatum*. Also, one (1) beach species was identified, namely, *Pongania pinnata*.

Table 3. Initial Data on Mangrove Forest Inventory in seven (7) LGUs

Common Name	Genus	Specific epithet
1. Miapi/Kuyapi	*Avicennia*	*rumphiana*
2. Bungalon	*Avicennia*	*marina*
3. Miapi/Kuyapi	*Avicennia*	*officinalis*
4. Pototan/Busai	*Bruguiera*	*cylindrica*
5. Hangarai	*Bruguiera*	*parviflora*
6. Baras-baras	*Ceriops*	*decandra*
7. Lipata/Buta-buta	*Excoecaria*	*agallocha*

8. Bakawang Babae	*Rhizophora*	*mucronata*
9. Bakawang Lalaki	*Rhizophora*	*apiculata*
10. Bakawang Bato	*Rhizophora*	*stylosa*
11. Pagatpat	*Sonneratia*	*alba*
12. Pedada	*Sonneratia*	*ovata*
13. Tabigi	*Xylocarpus*	*granatum*
14. Marukbarok/Bani (Beach species)	*Pongania*	*pinnata*

Photo documentation



Figure 3. Project Team from Partido State University - Tinambac Campus, coordinating with the Brgy Captain of New Calauan, in Tinambac, Camarines Sur on May 22, 2023, and mangrove and fisheries assessment in Tinambac mangrove sites.

Fisheries Assessment

Fisheries Assessment conducted in the Second Quarter of Year 2023. Below is the summarized table of fisheries assessment in seven LGUs.

Table 4. Second Quarter Fisheries Assessment

Identification of Fish and Macro-invertebrates associated in Mangrove Areas of San Miguel Bay	
English Name	Scientific name
Cockle	*Anadara spp.*
Kapis	*Isognomon sp.*
Talaba	*Crassostrea spp*
Orange fiddler	*Tubuca coarctata*
Mangrove tree crab	*Aratus pisonii*
Spotted moon crab	*Ashtoret lunaris*
Mud crab	*Scylla sp.*
Ring-legged crab	*Austruca annulipes*
Yellow-orange crab	
Shrimp	*Metapenaeus sp.*
Snapping shrimp	*Alpheus sp.*
Duckbill	*Linguna anatina*
Venus clam	*Chamelea sp.*
Manila clam	
Goby	
Halfbeak	
Hardy silverside	*Atherinomorus sp.*
Sea krait	
	Batillaria spp.
	Cerithidea spp.
	Hermit crab
	Nerita spp. 1
	Ocenabra sp.
	Potamides sp.
	Pirenella

	Austruca triangularis
	Gelasimus jocelynae

Photo Documentation



Figure 4. Project Team from Partido State University - Tinambac Campus, during the conduct of fisheries assessment.

Financial Status

The project was granted a total research fund of PHP 4,500,000.00 for two (2) years (September 16, 2022 to September 15, 2024). In this line, a total of 1,938,242.00 was requested for Year-2023 (January to December 2023) expenses. As of June 30, 2023, the estimated budget utilization during this year is PHP 558,035.71 (Table 5).

Table 5. Financial Status of the Mangrove Research Project of ParSU-Tinambac as of June 30, 2023, funded by CHED-GIA.

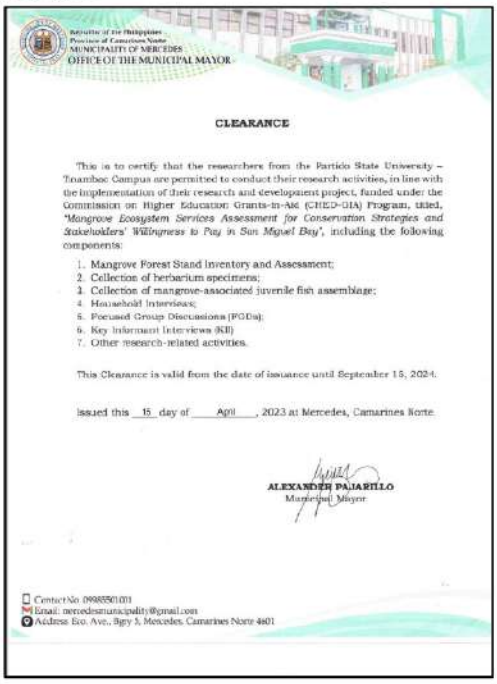
Partido State University		
Agency		
FUND UTILIZATION REPORT		
Project: Research Innovation and Extension Proposal: Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay		
As of June 30, 2023		
Beginning Balance	-	
Funds Received	1,670,984.00	**1,670,984.00**
Funds Utilized		
I.Personnel Services (PS)		
Personnel Services		
Honoraria (Project Leader & Members)	138,900.00	
Sub- Total		138,900.00
II. Maintenance and Other Operating Expenses (MOOE)		
Travel Expenses	82,000.00	
Supplies and Materials	139,300.38	
Communication Expenses	-	
Professional Services		
COS-Project Support Staffs	197,835.33	
Enumerators' Fee	-	
Other MOOE		
Printing and Publication Expenses	-	
Representation Expenses	-	
Subscription Expenses	-	
Sub-Total		419,135.71
III. Admin Cost (MOOE and Honoraria for Admin Support Staffs)		-
IV. Equipment Outlay		-
TOTAL		558,035.71
Ending Balance (Total Funds Received - Total Funds Utilized)		**1,112,948.29**

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APPENDICES

Appendix 1. Municipal Clearance

Appendix 1.1 Mercedes Clearance



Republic of the Philippines
Province of Camarines Norte
MUNICIPALITY OF MERCEDES
OFFICE OF THE MUNICIPAL MAYOR

CLEARANCE

This is to certify that the researchers from the Partido State University – Tinambac Campus are permitted to conduct their research activities, in line with the implementation of their research and development project, funded under the Commission on Higher Education Grants-in-Aid (CHED-GIA) Program, titled, *"Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay"*, including the following components:

1. Mangrove Forest Stand Inventory and Assessment;
2. Collection of herbarium specimens;
3. Collection of mangrove-associated juvenile fish assemblage;
4. Household Interviews;
5. Focused Group Discussions (FGDs);
6. Key Informant Interviews (KII)
7. Other research-related activities.

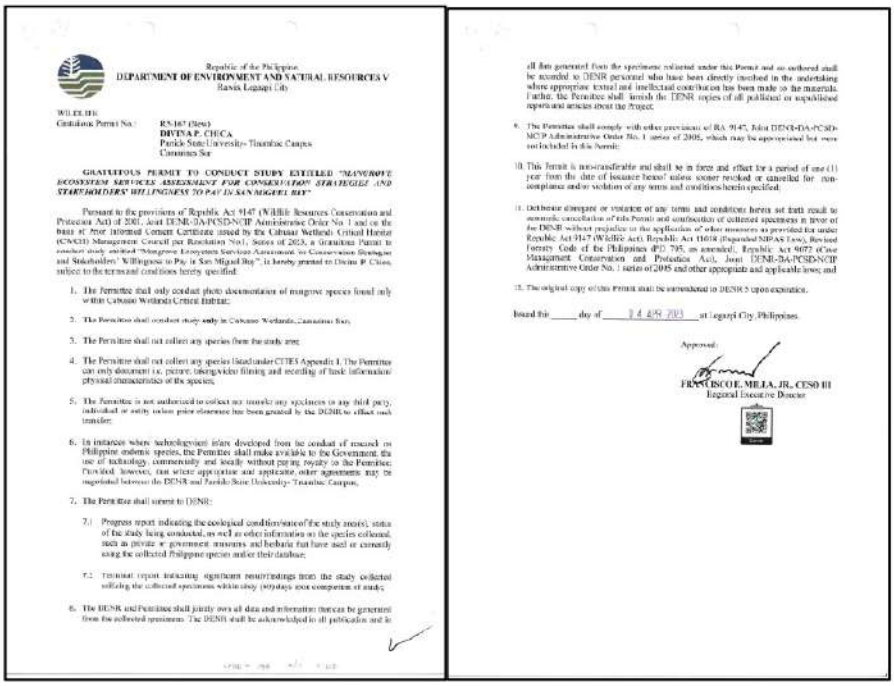
This Clearance is valid from the date of issuance until September 15, 2024.

Issued this 15 day of April, 2023 at Mercedes, Camarines Norte.

ALEXANDER PAJARILLO
Municipal Mayor

Contact No. 09985501001
Email: mercedesmunicipality@gmail.com
Address: Eco. Ave., Brgy 5, Mercedes, Camarines Norte 4601

Appendix 2. Gratuitous Permit for Cabusao Wetland Critical Habitat



Republic of the Philippine
DEPARTMENT OF ENVIRONMENT AND NATURAL RESOURCES V
Rawis Legazpi City

WILDLIFE
Gratuitous Permit No.: RS-167 (New)

DIVINA P. CHICA
Partido State University- Tinambac Campus
Camarines Sur

GRATUITOUS PERMIT TO CONDUCT STUDY ENTITLED *"MANGROVE ECOSYSTEM SERVICES ASSESSMENT FOR CONSERVATION STRATEGIES AND STAKEHOLDERS' WILLINGNESS TO PAY IN SAN MIGUEL BAY"*

Pursuant to the provisions of Republic Act 9147 (Wildlife Resources Conservation and Protection Act) of 2001, Joint DENR-DA-PCSD-NCIP Administrative Order No. 1 and on the basis of Prior Informed Consent Certificate issued by the Cabusao Wetlands Critical Habitat (CWCH) Management Council per Resolution No.1, Series of 2023, a Gratuitous Permit to conduct study entitled "Mangrove Ecosystem Services Assessment for Conservation Strategies and Stakeholders' Willingness to Pay in San Miguel Bay" is hereby granted to Divina P. Chica, subject to the terms and conditions hereby specified:

1. The Permittee shall only conduct photo documentation of mangrove species found only within Cabusao Wetlands Critical Habitat;
2. The Permittee shall conduct study only in Cabusao Wetlands, Camarines Sur;
3. The Permittee shall not collect any species from the study area;
4. The Permittee shall not collect any species listed under CITES Appendix I. The Permittee can only document i.e. picture, taking video filming and recording of basic information physical characteristics of the species;
5. The Permittee is not authorized to collect nor transfer any specimens to any third party, individual or entity unless prior clearance has been granted by the DENR to effect such transfer;
6. In instances where technology/ies is/are developed from the conduct of research on Philippine endemic species, the Permittee shall make available to the Government, the use of technology, commercially and locally without paying royalty to the Permittee; Provided, however, that where appropriate and applicable, other agreements may be negotiated between the DENR and Partido State University- Tinambac Campus;
7. The Permittee shall submit to DENR:
   7.1 Progress report indicating the ecological condition/status of the study area(s), status of the study being conducted, as well as other information on the species collected, such as private or government museums and herbaria that have used or currently using the collected Philippine species and/or their database;
   7.2 Terminal report indicating significant results/findings from the study collected reflecting the collected specimens within sixty (60) days upon completion of study;
8. The DENR and Permittee shall jointly own all data and information that can be generated from the collected specimens. The DENR shall be acknowledged in all publication and in all data generated from the specimens collected under this Permit and no authored shall be recorded to DENR personnel who have been directly involved in the undertaking where appropriate textual and intellectual contribution has been made to the materials. Further, the Permittee shall furnish the DENR copies of all published or unpublished reports and articles about the Project;
9. The Permittee shall comply with other provisions of RA 9147, Joint DENR-DA-PCSD-NCIP Administrative Order No. 1 series of 2005, which may be appropriate but were not included in this Permit;
10. This Permit is non-transferable and shall be in force and effect for a period of one (1) year from the date of issuance hereof unless sooner revoked or cancelled for non-compliance and/or violation of any terms and conditions herein specified;
11. Deliberate disregard or violation of any terms and conditions herein set forth result to automatic cancellation of this Permit and confiscation of collected specimens in favor of the DENR without prejudice to the application of other measures as provided for under Republic Act 9147 (Wildlife Act), Republic Act 11038 (Expanded NIPAS Law), Revised Forestry Code of the Philippines (PD 705, as amended), Republic Act 9072 (Clean Management Conservation and Protection Act), Joint DENR-DA-PCSD-NCIP Administrative Order No. 1 series of 2005 and other appropriate and applicable laws; and
12. The original copy of this Permit shall be surrendered to DENR 5 upon expiration.

Issued this ____ day of 24 APR 2023 at Legazpi City, Philippines.

Approved:

FRANCISCO E. MEJIA, JR., CESO III
Regional Executive Director